

IntaRNA profiling

Findings and proposed next steps for Martin

Tested revision: refactor-to-c++23 / 1b8377989ba7. This is a profile of one recorded build; no comparison with an earlier release or another PR was performed.

Recommendation

Fix the confirmed input-stream cleanup defect first. Then review the accessibility-file limit behavior and benchmark narrowly scoped CPU optimizations, separately for default, exact and ensemble modes.

Confirmed defect: input streams are not cleaned up

Reproduction: 100 open/read/cleanup cycles against the freshly built library increased open file descriptors from **4 to 104**. All four heap traces also retained 9,384 bytes in input-stream allocation stacks.

Cause: `src/IntaRNA/general.cpp:136` casts the input stream to `filtering_ostream`, although it was created as `filtering_istream`. The cast fails and cleanup is skipped. Resources accumulate in repeated library use; the OS reclaims them when a CLI process exits. Introduction history was not investigated.

Performance findings worth acting on

Finding	Measured evidence	Interpretation / boundary
Default X/H CPU	Loop-energy evaluation: 57.6% of 368 CPU samples in the biological GcvB batch.	First optimization target: repeated internal-loop energy work. Sample share is not a promised speedup.
Batch parallelism	48 targets × ChiX: 4.944 → 0.974 s at 1 → 12 threads (5.08×).	Peak RSS: 18.7 → 65.3 MiB. Batch throughput; not single-pair acceleration.
Accessibility reuse	0.805 → 0.492 s (39% lower latency); same base-pair list.	Matched target/query caps: 149/99. One-time generation: 0.847 s, excluded from cached-call time.
Windowing, S/H	127.5 → 24.1 MiB (-81%); 21.38 → 59.71 s (2.79× time).	3,000 × 800 nt; window 300, overlap 150. Default X/H gets no useful RSS reduction and takes 1.72× longer.

Proposed sequence

1. Isolate the cleanup fix and add repeated plain/gzip stream tests. **2.** Confirm intended ED-file limit semantics with Martin. **3.** Agree on a production workload, then measure an energy/scoring optimization against this baseline. Proposed changes have not been implemented or assigned an expected speedup.

How to read the results

Reproducible baseline

56 configurations: 168 measured repetitions plus 56 excluded warmups (224 CLI runs). Three measured repetitions per case, deterministic randomized order, sequential execution. Medians are shown; full ranges and raw records are supplied. CPU sampling, heap tracing and diagnostic logging were separate from clean timings.

Host/build: Ryzen 5 7530U (6 cores / 12 hardware threads), Linux 6.8.0-139; GCC 16.1.0, C++23, -O3 with symbols/frame pointers; ViennaRNA 2.7.2, Boost 1.85.0. OPENBLAS_NUM_THREADS=1, OMP_DYNAMIC=FALSE, OMP_PLACES=cores and OMP_PROC_BIND=close. Exact commands, hashes, dependencies and input sequences accompany the report.

Batch scaling: choosing a thread count

Threads	Median wall s	Observed range s	Peak RSS MiB	Speedup
1	4.944	4.929–5.006	18.7	1.00×
2	2.556	2.527–2.557	23.0	1.93×
4	1.457	1.441–1.465	31.4	3.39×
6	1.104	1.103–1.130	40.0	4.48×
12	0.974	0.970–0.997	65.3	5.08×

48 evaluated 300-nt targets × ChiX (84 nt); 46 reportable interactions. Six threads supply 88% of twelve-thread throughput at 61% of its RSS. Peak RSS values are medians of per-process high-water marks.

Two interpretation traps

Cached ED is not automatically a transparent replacement. The writer emits columns through its maximum; the reader advertises one nucleotide less for dangling-end treatment (AccessibilityFromStream.cpp:90). We therefore compared computed and loaded ED at the same 149/99 limits. Whether this is intended needs review before generalizing the saving.

Mode changes are not implementation speedups. Exact X/M is dominated by repeated scoring terms (Boltzmann weights 19.4%, ED lookup 18.5% of samples); ensemble mode adds partition-function recurrence and map costs. Disabling accessibility changes the scientific model and was slower in one test. Windowing and algorithm substitutions need their own output checks.

Validation and uncertainty

All timed cases and the automated evidence audit passed; 4,281 assertions in 36 API tests and all 20 CLI regression cases passed. Base-pair lists matched for 1 vs 12 threads and the matched ED comparison. Window comparisons matched coordinates and energy; full structure equality was not separately checked there.

This is one laptop, three measured repetitions per case and a bounded workload suite. No release-wide regression or biological-accuracy claim follows. CPU profiles total 1,182 samples at 100 ms. A collector timer warning and incomplete optimized stack ancestry limit precision, although CPU totals were consistent. Hardware counters were blocked; no cache-miss or IPC claims are made.

Evidence and reporting conventions

Attached evidence archive: start with **profiling/REPORT.md**. Inspect **results/summary.csv**, **results/audit.json**, **results/stream-cleanup-probe.json**, **cases.json**, and the reproduction scripts. Original data and hashes are retained.

Reporting follows full configuration disclosure and explicit warmup/repetition statistics, as illustrated by **SPEC reporting rules** and the **Google Benchmark guide**. This custom IntaRNA study does not claim SPEC certification.